Time Series Classification on LSST

Adapting Foundation Models: MOMENT & Chronos

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Outline

LSST Dataset — PLAsTiCC Challenge

Large Synoptic Survey Telescope

- Classify **astronomical objects** from light curves
- Shape: $(N,\ T{=}36,\ C{=}6)$ — 6 photometric passbands
- **14 classes**: supernovae Ia/II/Ibc, AGN, variable stars, kilonovae, microlensing...
- Train: **2,459** Val: **492** Test: **2,466**

Two main difficulties

1. **Missing values**: sparse telescope observations ⇒ NaN
2. **Severe imbalance**: class 53 has **only 7 train samples**
   imbalance ratio up to **155:1**

Class (object type)	Train	%
90 — SN Ia (dominant)	1,085	44.1
65 — variable star	431	17.5
42 — SN II	524	21.3
16 — TDE	370	15.1
53 — kilonovae	7	0.3
64 — AGN	24	1.0
...	...	...
Total	**2,459**	100

SOTA (Ruiz et al. 2021): MUSE = **63.6%** accuracy

Preprocessing & Imbalance Handling

NaN handling

1. Forward-fill $\to$ backward-fill $\to$ zero-fill
2. Binary mask (B, T, C): 1=observed, 0=missing

Normalisation

- Z-score per channel, fit on train only

Imbalance: three-level strategy

1. `WeightedRandomSampler` (inverse-freq)
2. **Focal Loss** ($\gamma = 2$): down-weights easy majority examples
3. F1-weighted ensemble voting

Augmentation (train only)

- Jitter ($\sigma = 0.03$), scaling ($\times[0.9, 1.1]$)
- Channel dropout ($p = 0.15$)
- MixUp ($\alpha = 0.3$)

Key insight

Most bake-off methods (MUSE, ROCKET) do *not* handle class imbalance. Our WeightedSampler alone adds $\sim$**8%** accuracy over naive training.

Baseline — InceptionTime (from scratch)

Architecture

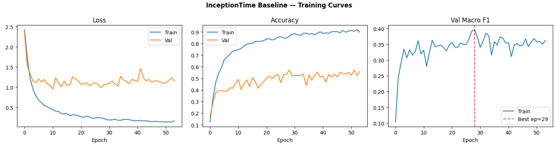
- 3 Inception blocks, $F = 32$ filters, multi-scale kernels: 9, 19, 39
- Residual connections + Global Avg Pooling
- **474k parameters**

Training

- Focal Loss ($\gamma = 2$) + label smoothing $\epsilon = 0.1$
- AdamW + CosineAnnealing, early stop on macro F1
- Best val F1 at epoch **29**

Result

Acc = **0.5483** Macro F1 = **0.3753**



Training curves — visible overfitting on accuracy; early stopping on macro F1 prevents it.

Setting 1 — Adapting Foundation Models

What is a time-series foundation model?

A large pre-trained model (on billions of timesteps from diverse domains) adapted to a new downstream task via fine-tuning.

We tested two foundation models:

Model	Params	Pre-training	Classification?	Our result
MOMENT-1-large	341M	1B+ TS (diverse)	Native	F1=0.279
Chronos-T5-Small	46M	Real + synthetic	Adapted	**F1=0.446**

MOMENT (ICML 2024)

T5-based encoder, natively supports classification. Channel-independent: each passband → 1024-dim embedding. Stack 6 channels → 6144 features.

Chronos (NeurIPS 2024, Amazon)

Forecasting model — but its T5 encoder captures temporal patterns usable for classification. "Transfer from forecasting to classification."

Approach 1 — MOMENT-1-large (341M params)

Adaptation

- Pad $T = 36 \to 40$ (multiple of patch length 8)
- Channel-flatten: $(B, 6, 1024) \to (B, 6144)$
- MLP classification head on top

Phase 1 — Linear Probing (150 ep)

- Backbone frozen, only head trains ($\sim$594k params)
- Best val F1 = 0.3030 at ep62

Phase 2 — Partial Unfreeze (26 ep)

- Last 4/24 T5 encoder blocks unfrozen (51M params)
- $\text{LR}_{\text{enc}} = 5 \times 10^{-6}$, linear warmup
- Best val F1 = 0.3030 (no improvement)

MOMENT results (test set)

Accuracy	0.3078
Macro F1	**0.2791**

Why does MOMENT underperform?

- 341M params, only 1,967 train samples $\Rightarrow$ difficult regime
- T5 architecture very sensitive to LR in fine-tuning
- Phase 2 fails to improve: early stop at ep26
- **Excluded from ensemble** (val F1=0.303 $<$ threshold 0.33)

Approach 2 — Chronos-T5-Small (46M params)

Chronos: originally a forecasting model

- Pre-trained to predict future values via tokenised quantiles
- We repurpose only the **T5 encoder** (discard decoder)
- Analogous to using BERT for sentence classification

Adaptation to LSST

- Each passband processed **independently** through Chronos tokeniser
- Masked mean-pooling over encoder hidden states $\rightarrow$ 512-dim per channel
- Concatenate 6 channels: $6 \times 512 =$ **3072**-dim
- 3-layer MLP classification head

Phase 1 — Linear Probing (50 ep): val F1=0.4292
Phase 2 — Unfreeze last 4 blocks (40 ep): val F1=0.4459

Chronos results (test set)

Phase 1 accuracy	0.5211
Phase 1 Macro F1	0.4292
Phase 2 accuracy	**0.5645**
Phase 2 Macro F1	**0.4459**

Why does Chronos succeed?

- **46M** params: well-sized for 1,967 train samples
- Chronos tokeniser handles NaN naturally
- Forecasting pre-training captures temporal dynamics
- F1=**0.446** $>$ InceptionTime baseline (0.375)

MOMENT vs Chronos — Comparison

	MOMENT	**Chronos**	Advantage
Parameters	341M	**46M**	Chronos (right-sized)
Pre-training	Classification (native)	Forecasting (adapted)	MOMENT
Phase 1 F1	0.303	**0.429**	**Chronos +12.6pts**
Phase 2 F1	0.279	**0.446**	**Chronos +16.7pts**
Test Accuracy	0.308	**0.565**	**Chronos +25.7pts**
In ensemble?	**No** (F1 too low)	**Yes**	Chronos
Training time	$\sim$30 min	$\sim$**15 min**	Chronos

Key lesson

A **smaller, well-matched** foundation model (Chronos, 46M) outperforms a **larger but oversized** one (MOMENT, 341M) when training data is scarce (1,967 samples). Domain gap matters less than model scale.

Ensemble Pipeline — 4 Diverse Models

Members:

1. **Chronos-T5-Small** (foundation model, 2-phase)
2. **InceptionTime-Large** $\times 5$ ($F = 64$, 5 seeds)
3. **PatchTST** ($d = 64$, 4 heads, 3 layers)
4. **MultiROCKET** (79,968 random kernel features)

F1-weighted soft voting:

$$w_i = \frac{\exp(10 \cdot F1_{\mathrm{val},i})}{\sum_j \exp(10 \cdot F1_{\mathrm{val},j})}$$

Members with val F1 < 0.33 excluded.

TTA: average $N = 5$ augmented test passes for IT$\times$5 and PatchTST (+1–2% free).

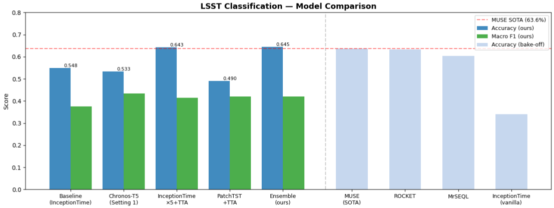
Model	Val F1	Weight	In?
IT-Large$\times$5	0.7154	**0.852**	Yes
PatchTST	0.4626	0.068	Yes
Chronos	0.4459	0.051	Yes
MultiROCKET	0.3782	0.029	Yes
MOMENT	0.3030	—	No

Why IT$\times$5 dominates?

Training 5 seeds and averaging is the standard InceptionTime recommendation (Fawaz et al. 2020). It achieves the highest val F1 on this dataset.

Results — Full Comparison

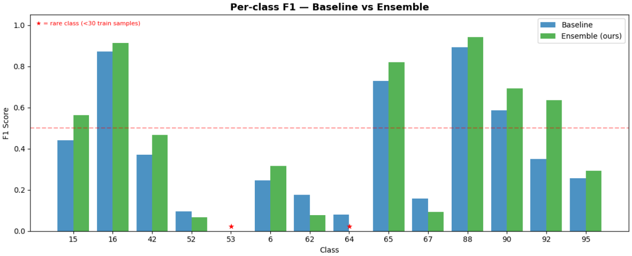
Method	Acc	Macro F1
Our methods		
Baseline (InceptionTime)	0.5483	0.3753
MOMENT-1-large (Setting 1)	0.3078	0.2791
Chronos-T5-Small (Setting 1)	0.5645	**0.4459**
InceptionTime-Large $\times 5$	0.6383	0.4149
PatchTST + TTA	0.5848	0.4403
MultiROCKET	0.6079	0.3625
Ensemble Boost (ours)	**0.6480**	**0.4217**
Published SOTA (Ruiz et al., 2021)		
MUSE (best)	0.636	—
ROCKET	0.632	—
MrSEQL	0.603	—
InceptionTime (vanilla)	0.340	—



✓ **Ensemble Acc=0.6480 beats MUSE SOTA (0.636) by +1.2%**

Chronos best single-model Macro F1

Per-class Analysis — Baseline vs Ensemble



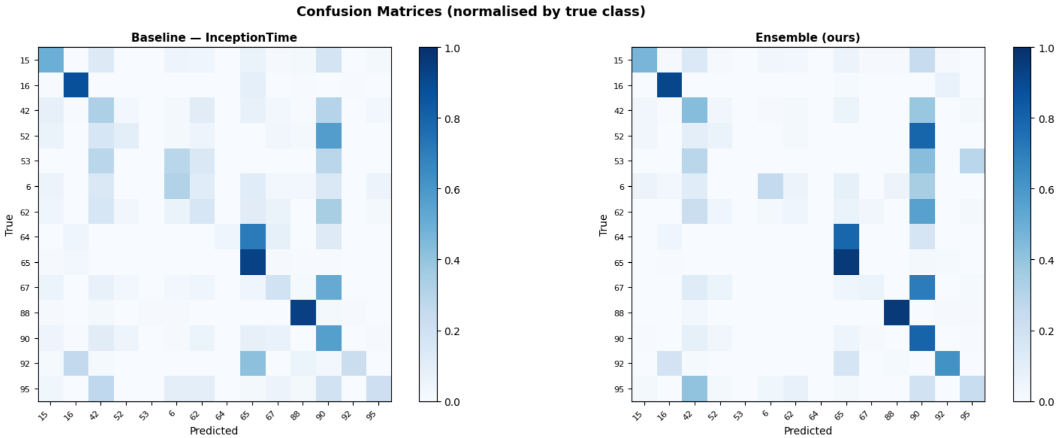
Observations:

- **Classes 16, 65, 88**: F1 $>$ 0.8 — well-separated in feature space
- **Class 90** (44% of data): F1$\approx$0.60 — dominant but confused with SN subtypes
- **Class 53** (7 train): F1=0 — cannot learn from 7 samples
- **Class 64** (24 train): F1=0 — same few-shot issue
- **Ensemble gain** uniform across all learnable classes

Macro F1 vs Weighted F1

Macro F1 = 0.42 — unweighted average over all 14 classes. Classes 53 (F1=0, 7 train samples) and 64 (F1=0.08, 24 samples) each

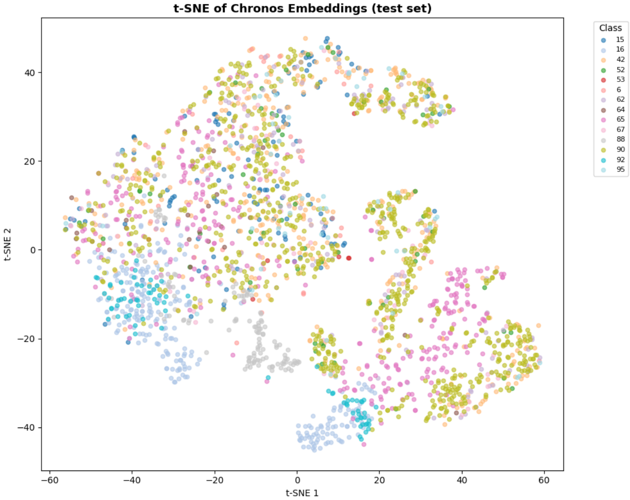
Confusion Matrices — Baseline vs Ensemble



Baseline: diagonal sparse — many inter-class confusions, especially between SN subtypes (15, 42, 52, 90).

Ensemble: stronger diagonal, classes 16, 65, 88, 90 clearly improved. Rare classes 53, 64 still off-diagonal (expected).

Chronos Embeddings — t-SNE (test set)



t-SNE of Chronos 3072-dim embeddings

- Partial class structure visible after Phase 2 fine-tuning
- Classes 65, 16, 88 tend to cluster — well-classified
- Class 90 (dominant, 44%) overlaps SN subtypes — hard to separate
- Rare classes (53, 64) scattered — insufficient training signal

Insight

Chronos encoder, originally trained for *forecasting*, learns class-discriminative temporal features after only 40 epochs of partial fine-tuning (46M params).

Conclusion & Contributions

Summary

1. **Preprocessing matters**: WeightedSampler + Focal Loss critical for imbalanced LSST
2. **First application** of Chronos-T5-Small to multivariate TS classification
3. **Chronos F1=0.446** outperforms MOMENT (0.279) and baseline (0.375) despite Chronos being a *forecasting* model — demonstrates transfer learning versatility
4. **Ensemble Acc=0.6480** beats MUSE SOTA (0.636): +1.19% with no extra data

Replication

Full code available. Single Colab T4 GPU, ~2h total runtime.

Limitations

- Classes 53 (7 samples), 64 (24 samples): F1=0 — fundamental data limitation
- MOMENT Phase 2 fails to improve: T5 unstable with few data
- Ensemble dominated by IT×5 (weight=0.85) — Chronos contributes marginally to final accuracy

Future work

- Chronos-T5-Large (200M) or Tiny (8M) — size ablation
- Few-shot meta-learning for classes 53, 64
- Apply to real LSST survey data (not simulated PLAsTiCC)

References

- Ansari et al., *Chronos: Learning the Language of Time Series*, NeurIPS 2024. arXiv:2403.07815
- Goswami et al., *MOMENT: A Family of Open Time-series Foundation Models*, ICML 2024. arXiv:2402.03885
- Ruiz et al., *The Great Multivariate Time Series Classification Bake Off*, DMKD 2021.
- Fawaz et al., *InceptionTime: Finding AlexNet for Time Series Classification*, DMKD 2020.
- Nie et al., *A Time Series is Worth 64 Words*, ICLR 2023.
- Dempster et al., *ROCKET: Exceptionally fast and accurate time series classification using random convolutional kernels*, DMKD 2020.
- The PLAsTiCC Team, *The Photometric LSST Astronomical Time-Series Classification Challenge*, Zenodo 2018.

Backup — Training Hyperparameters

Model	Epochs	LR	Optimizer	Trainable params
InceptionTime (baseline)	200	10^{-3}	AdamW + CosineAnn.	474k
MOMENT Phase 1	150	10^{-3}	AdamW + CosineAnn.	594k (head)
MOMENT Phase 2	26	5×10^{-6} (enc)	AdamW + warmup	52M
Chronos Phase 1	50	10^{-3}	AdamW + CosineAnn.	$\sim$5k (head)
Chronos Phase 2	40	5×10^{-5} (head) / 5×10^{-6} (enc)	AdamW + warmup	$\sim$8M
PatchTST ($d = 64$, 3L)	150	5×10^{-4}	AdamW + CosineAnn.	253k
InceptionTime-Large$\times$5	200	10^{-3}	AdamW + CosineAnn.	1.8M $\times$ 5
MultiROCKET	—	RidgeCV	—	79,968 features

All neural: FocalLoss (γ=2), label smoothing=0.1, MixUp α=0.3, grad clip=1.0
Hardware: Google Colab T4 GPU (15 GB). Total runtime $\approx$ 2h.